



DRONE MATRIX

THE FUTURE OF DRONE FLEET MANAGEMENT . . .

- **COLLEGE NAME :** UNITEDWORLD INSTITUTE OF TECHNOLOGY (UIT,KU)
- **TRL-2 PROOF OF CONCEPT :** TECHNOLOGY FORMULATED

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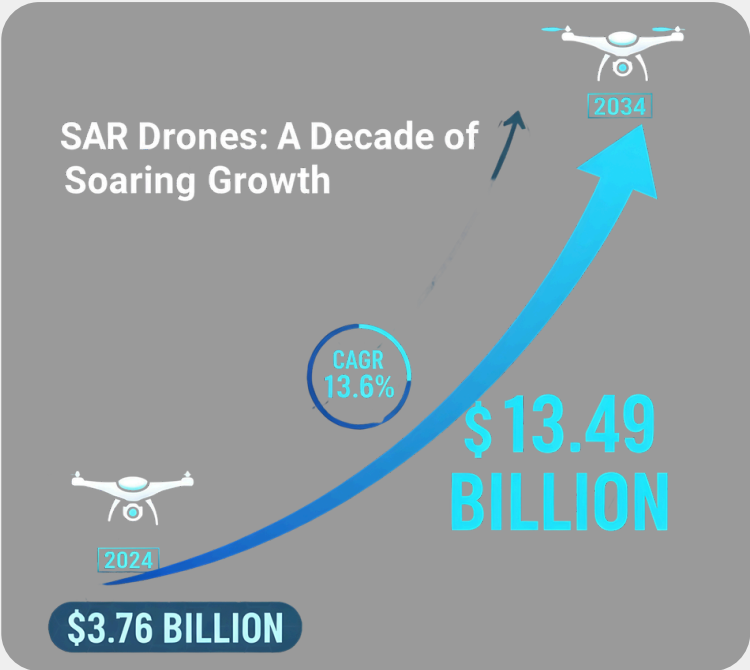
- **COURSE/SEM :** B.TECH[AIML], SEM:5
- **CITY :** GANDHINAGAR ; **STATE :** GUJARAT

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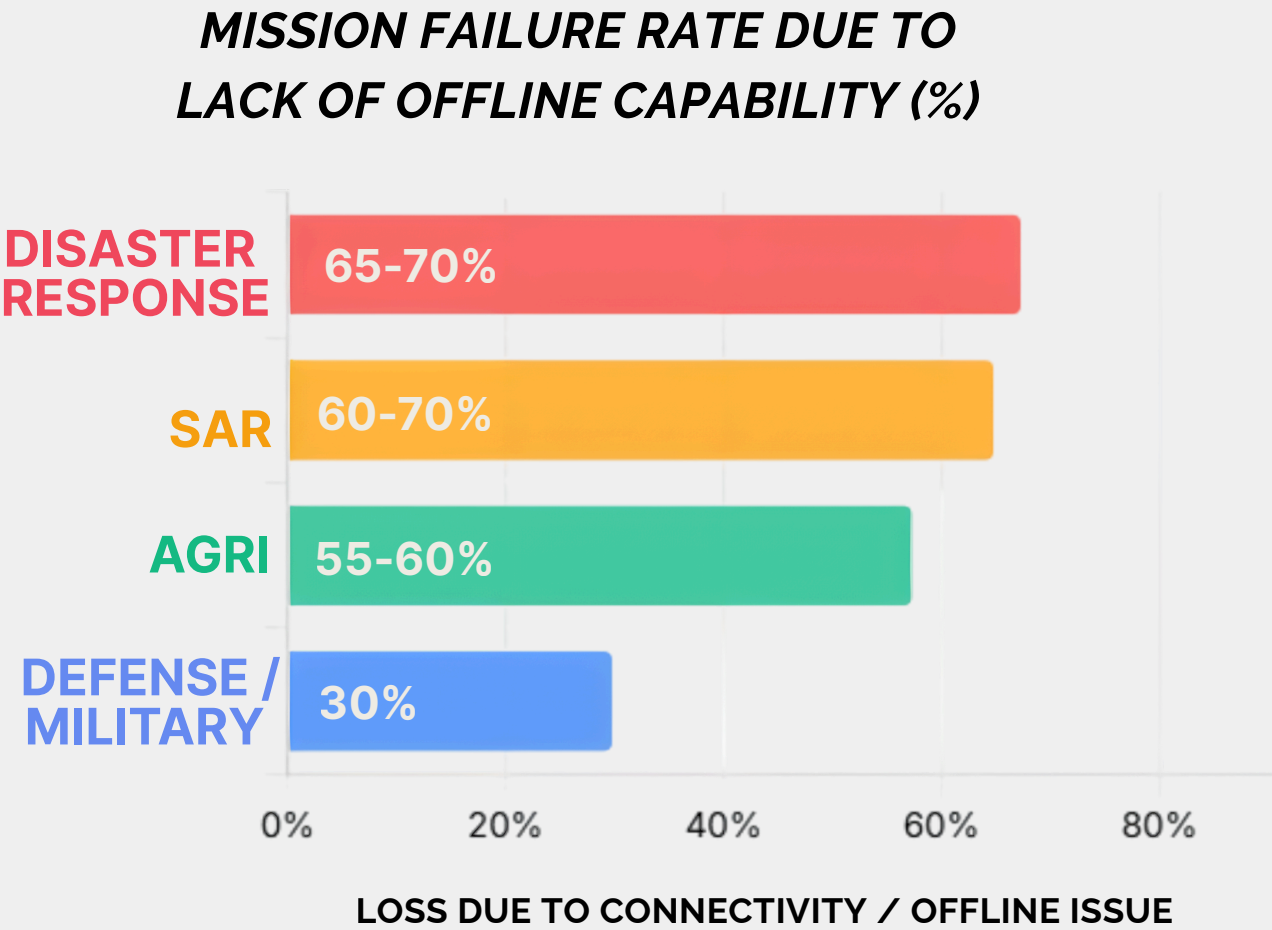
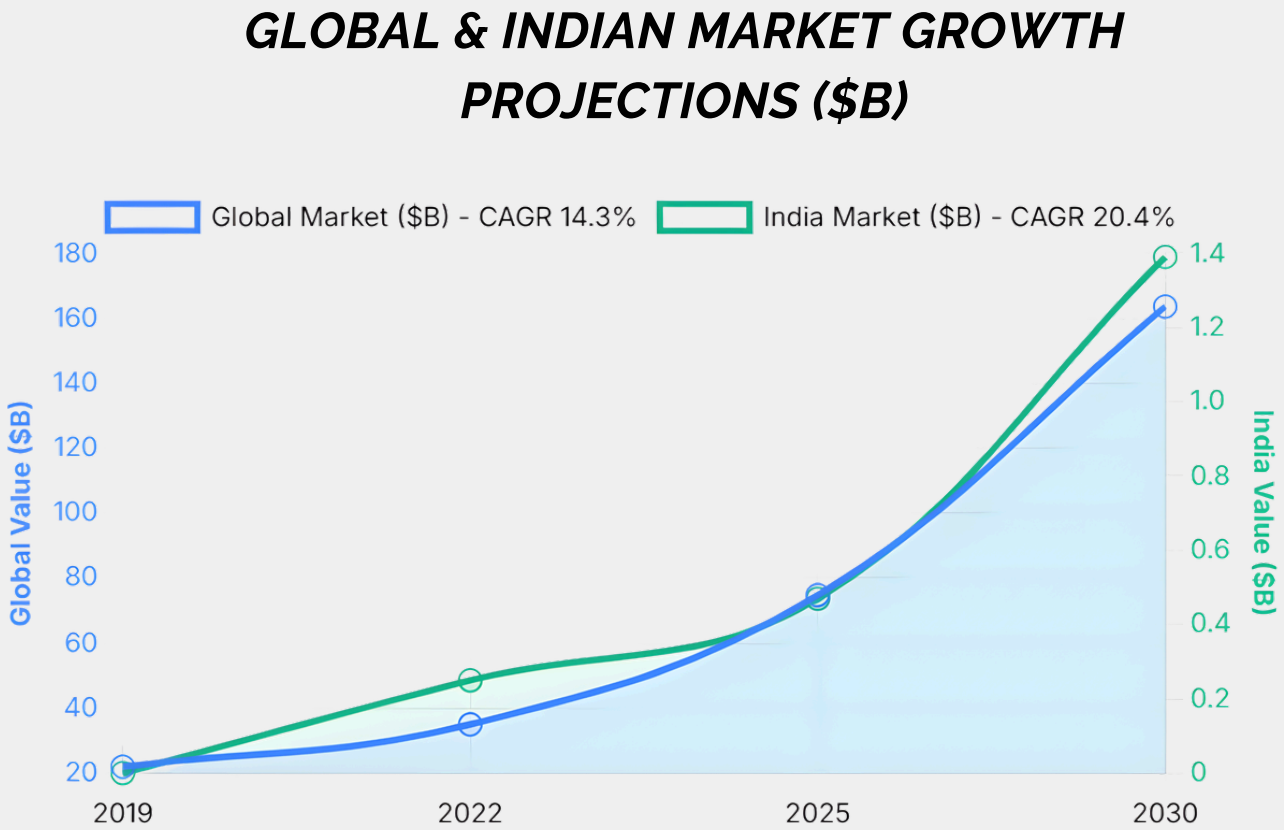
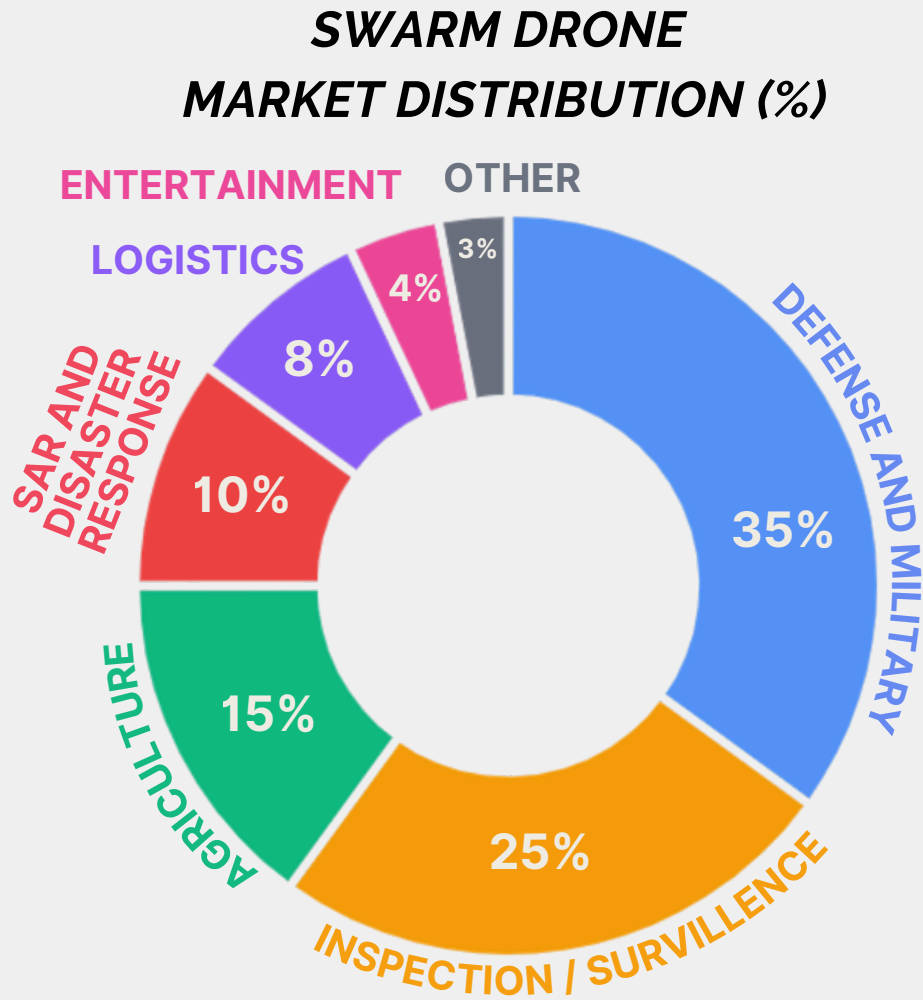
PROBLEM OVERVIEW

Critical drone missions fail not because of the drone, but because of unreliable connectivity and decentralized coordination, highlighting urgent need for robust, offline swarm coordination that works everywhere.

60%	SAR MISSIONS LOSE TELEMETRY/CONTROL
40-60%	RURAL GPS ACCURACY DROP (FORESTS/HILLS)



31%	RURAL CONNECTIONS MAINTAIN USABLE DATA SPEEDS
60%+	INDIAN RURAL REGIONS LACK RELIABLE TRACKING/INTERNET

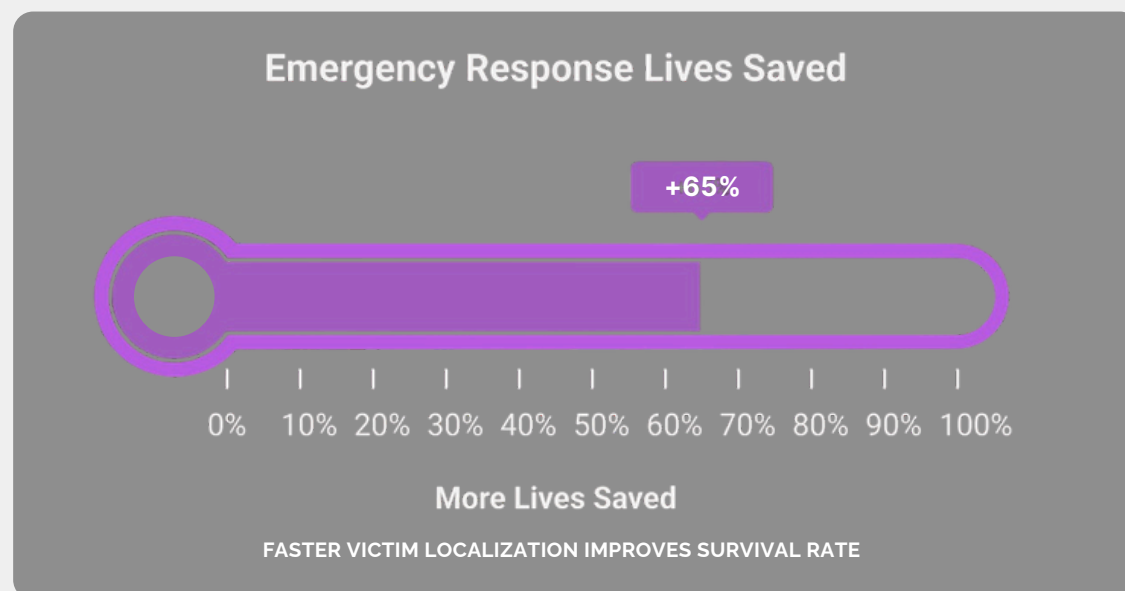
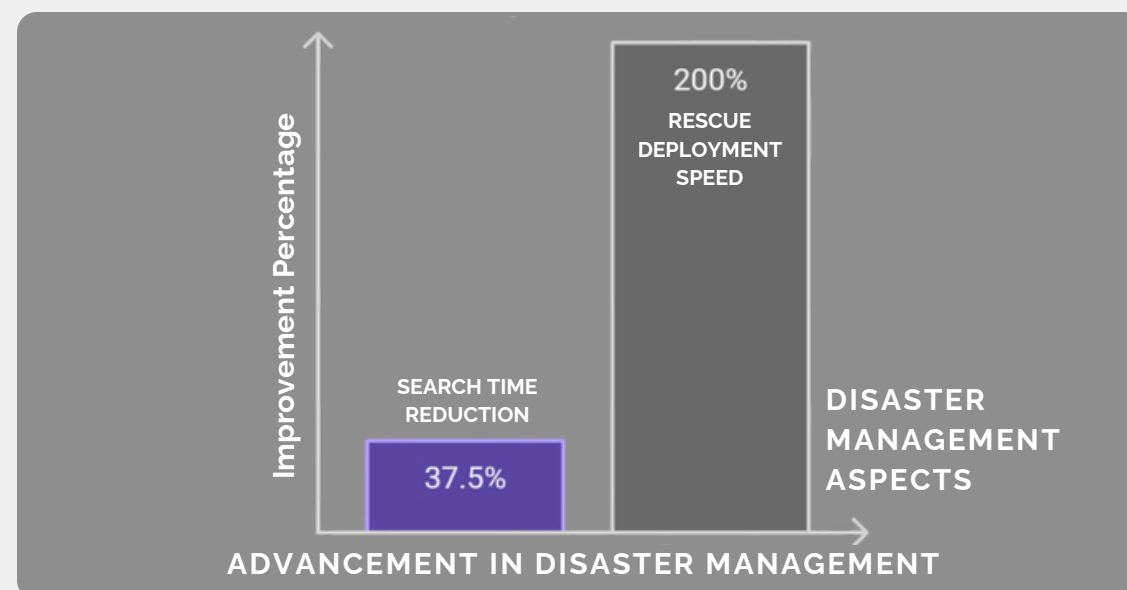
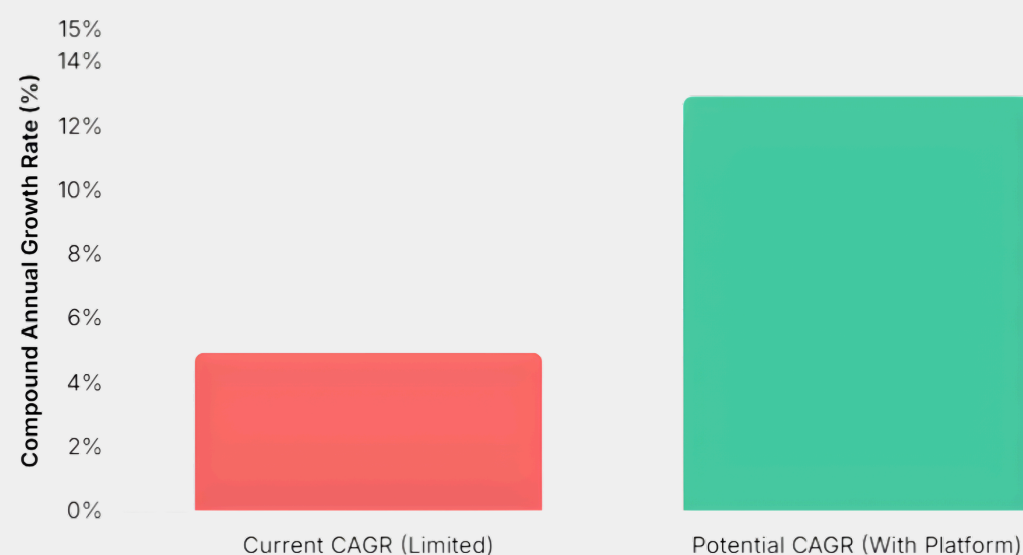


SOURCES: IEEE XPLORE (2022), ISRO NAVIC ACCURACY REPORT (2021), TRAI RURAL CONNECTIVITY REPORT (2023), DOT BHARATNET STATUS (2024), GRAND VIEW RESEARCH (2024-2034), MARKETSANDMARKETS DRONE SWARM REPORT (2024), DRDO FIELD DEPLOYMENT REVIEW (2022), INDIABIOSCIENCE, DATAM INTELLIGENCE

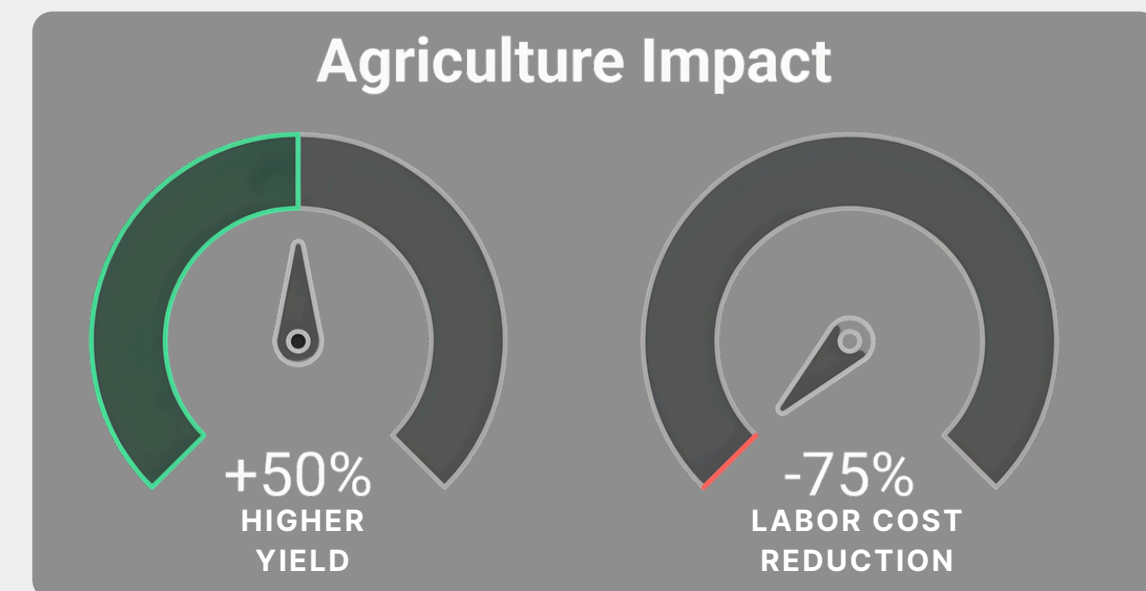
OUR SOLUTION

Drone Matrix is a plug and play, drone **swarm control software that lets one operator control and automate multi drone missions** anywhere, no cloud, decentralized, delivering ultra fast, resilient, AI driven coordination. Drone Matrix scales instantly without new hardware, reducing manpower and mission time significantly while improving reliability in agriculture, disaster response, and surveillance.

 **Swarm Market Growth: Current vs. Accelerated CAGR**



MILITARY AND DEFENSE



WORKFLOW

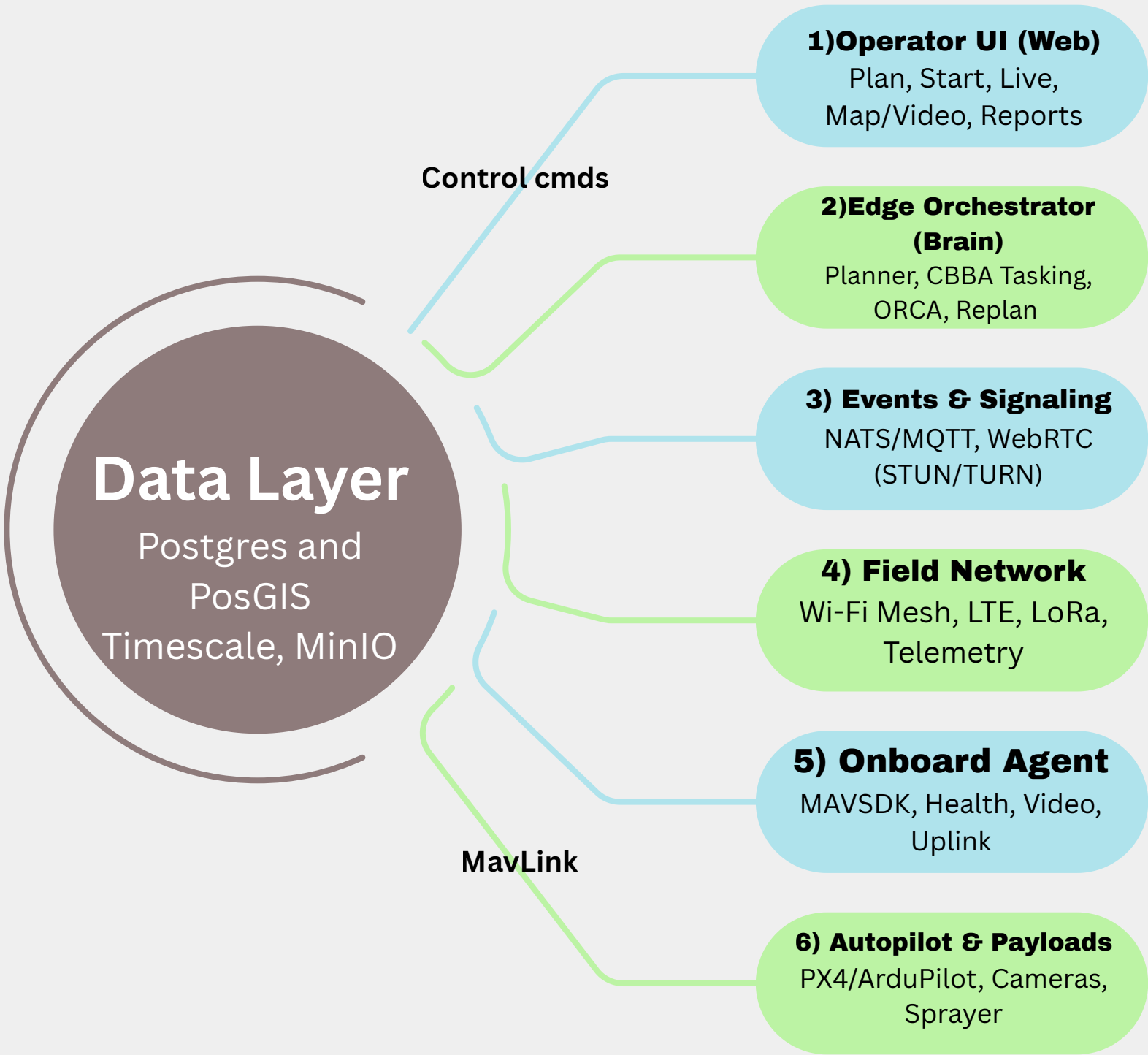
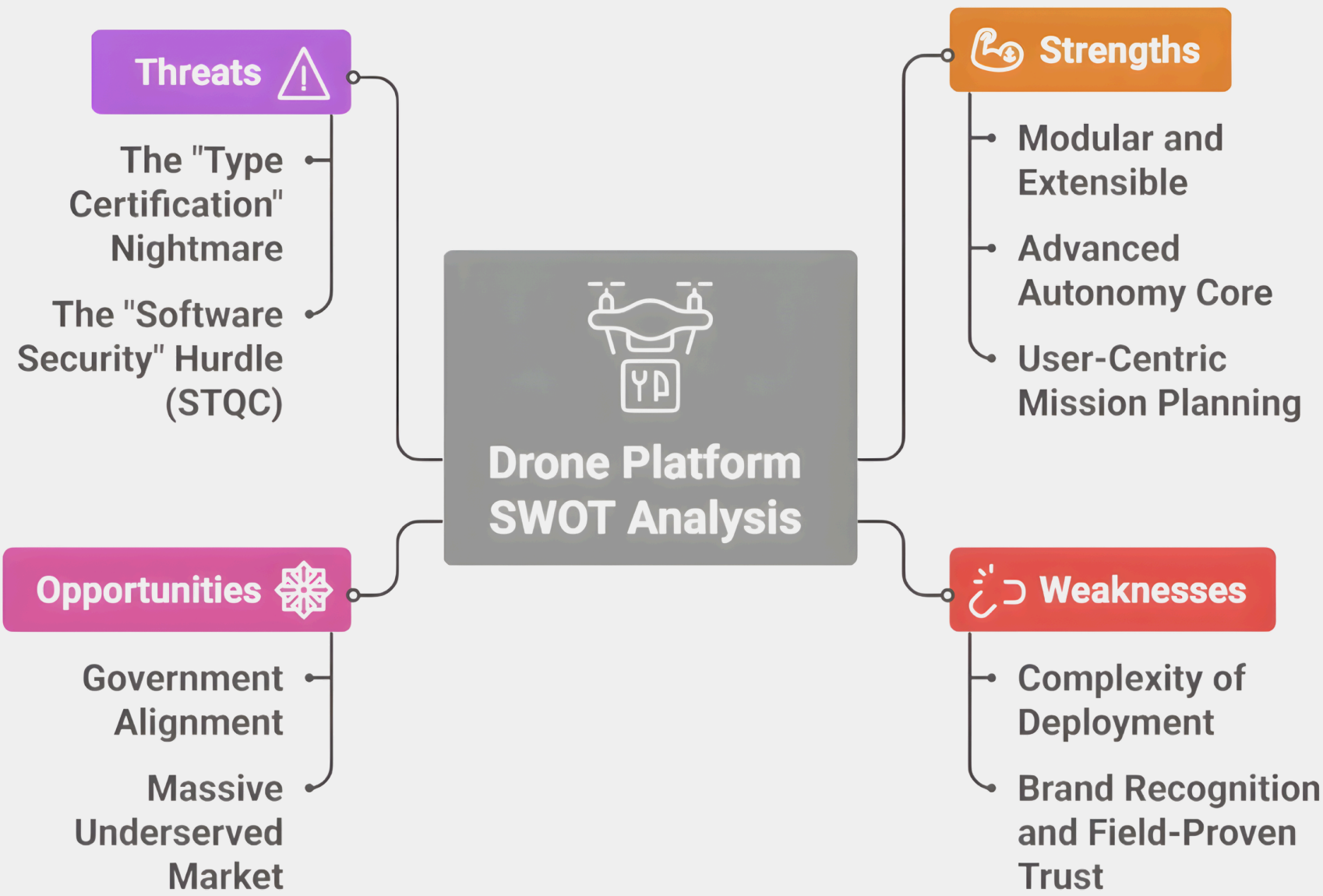


WORKFLOW



DRONE MATRIX SWOT ANALYSIS

TECH ARCHITECTURE



SWARM CO-ORDINATION AUTONOMY

CBBA: THE FAIR AUCTION FOR DRONE TEAMS

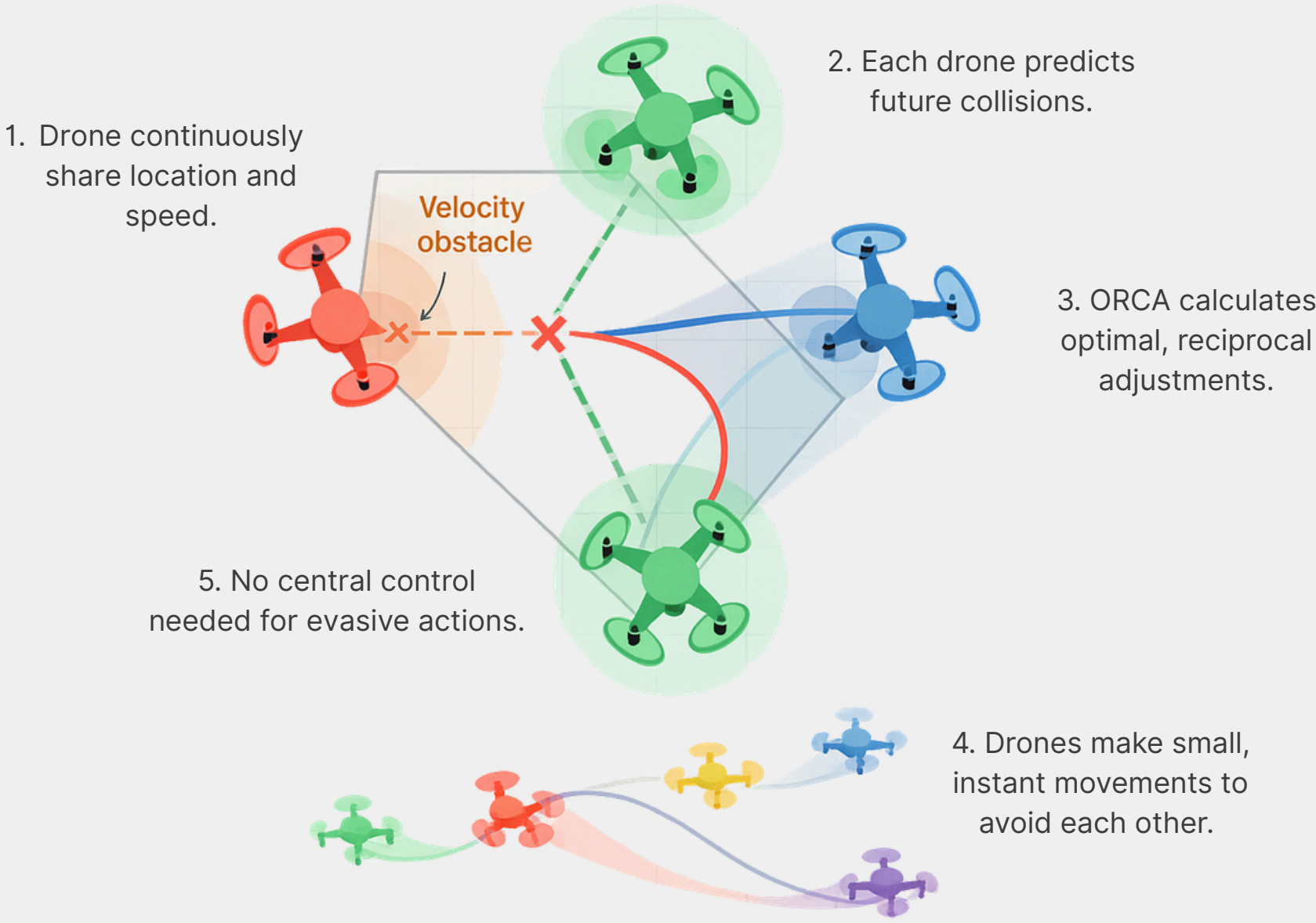
How Drone MatrX intelligently assigns tasks:



CBBA: FAST, DISTRIBUTED AND ADAPTS WELL, WHEN THINGS CHANGE IN THE FIELD

ORCA: THE INSTANT COLLISION AVOIDANCE SYSTEM

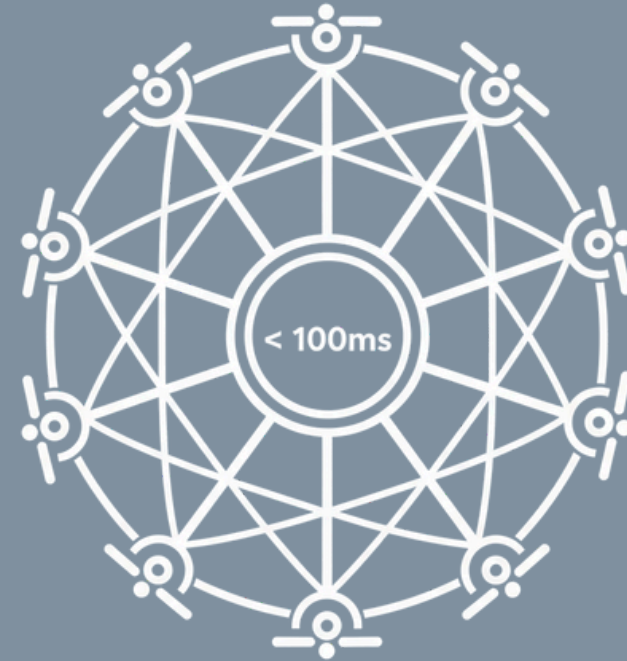
How Drone MatrX guarantees safety in dense drone operations:



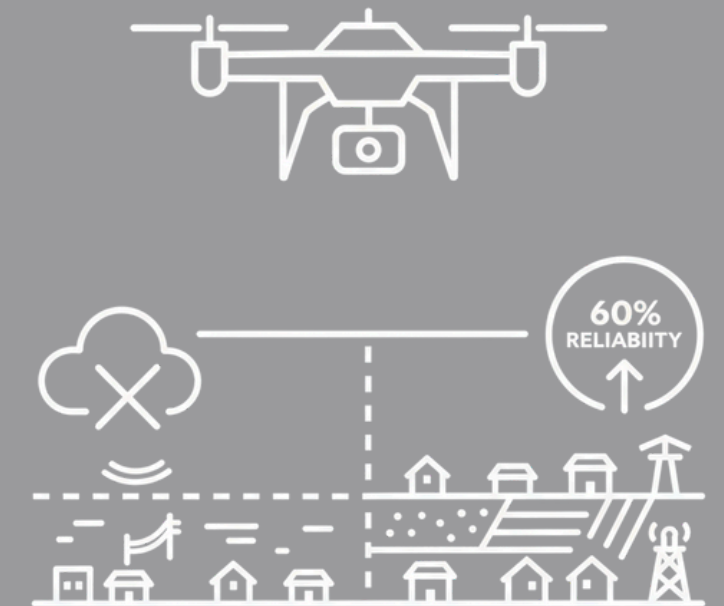
ORCA: ENSURES REAL-TIME SAFETY, MINIMIZES MISSION DISRUPTION, AND SCALES WITH FLEET SIZE.

UNIQUE SELLING PROPOSITION

Onboard communication cuts swarm control latency to <100ms.



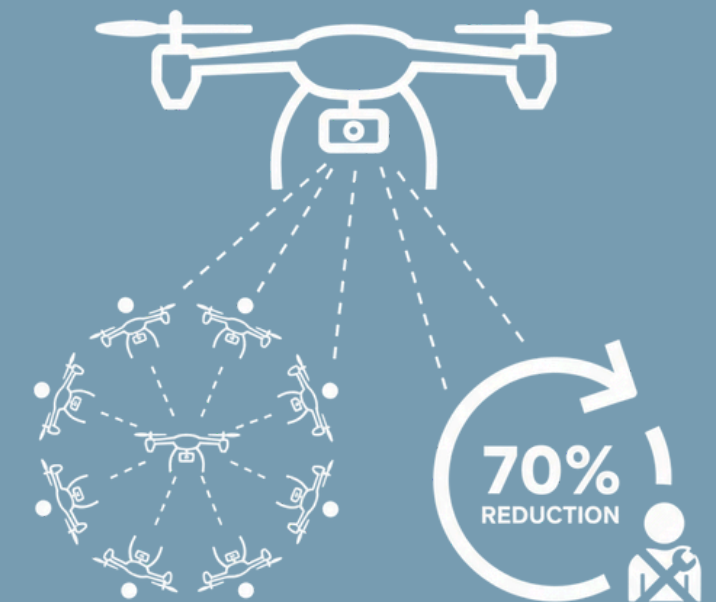
Cloud Independence (mission reliability increases in 60% rural & remote zones)



Plug and Play with Indian Drone Ecosystem (~80% of India's drone fleet using PX4, ArduPilot, and MAVLink standards)



Scalable Architecture for Faster Response (Automates task division and coordination, cutting manual workload by 70%)



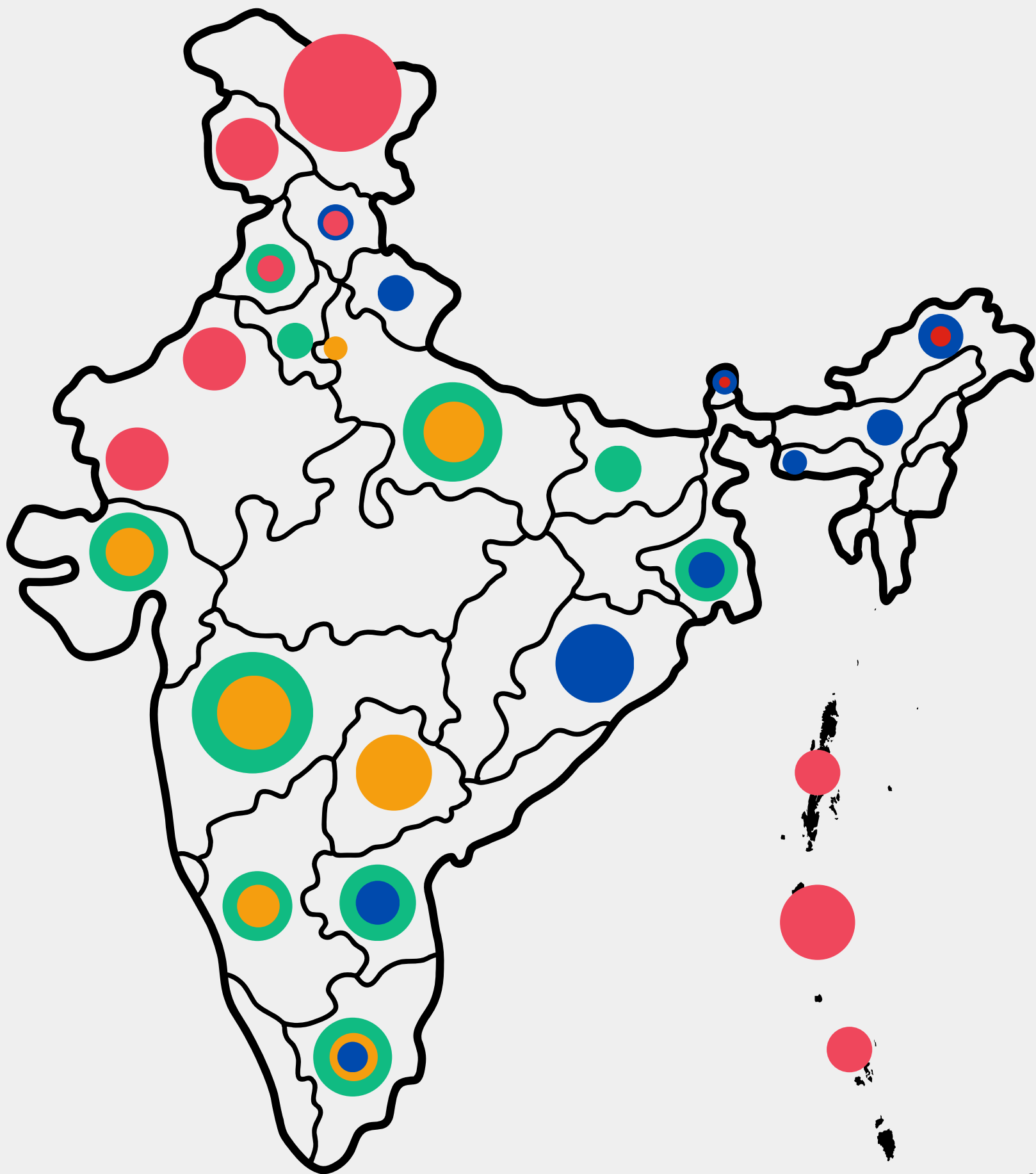


COMPETITIVE LANDSCAPE: OUR EDGE IN SWARM INTELLIGENCE

Particulars	DJI Dock	FlytBase	Auterion	UgCS	DroneDeploy	DARPA OFFSET	Drone Matrx
Feature 1: Autonomous Swarm Coordination	Partial (Single Drone)	Partial (≤10 drones)	Yes	No (Manual mission setup)	No	No	Full multi drone swarm autonomy ✓
Feature 2: AI-Based Mission Planning	Limited	Yes	Yes	Yes	Yes	No	AI-driven adaptive mission planning ✓
Feature 3: Real-Time Object Detection	Yes	Yes	Yes	No	Yes	Yes	Onboard + Cloud AI detection ✓
Feature 4: Processing Architecture	Cloud Only	Cloud-Based	Cloud Based	Local	Cloud	Edge + Cloud	Edge ✓
Feature 5: Open Architecture, Customizable	Closed	Semi-Open	Closed	Closed	Closed	Open Research Platform	Open-core modular architecture ✓
Feature 6: Offline Mode	No	No	No	Yes	No	Yes	Yes (local ops possible) ✓
Feature 7: Suitability	Moderate	Good	Limited	Limited	Moderate	R&D Only	Optimized for Govt, R&D, Startups ✓

Sources: Vendor and Product Websites (DJI Sources: Vendor Product Documentation (DJI, FlytBase, Auterion, UgCS, DroneDeploy); Defense and Research Publications (DARPA OFFSET Program); Industry Analyst/Market Research Reports on Drone Autonomy, Swarm Coordination, and UAS Software.

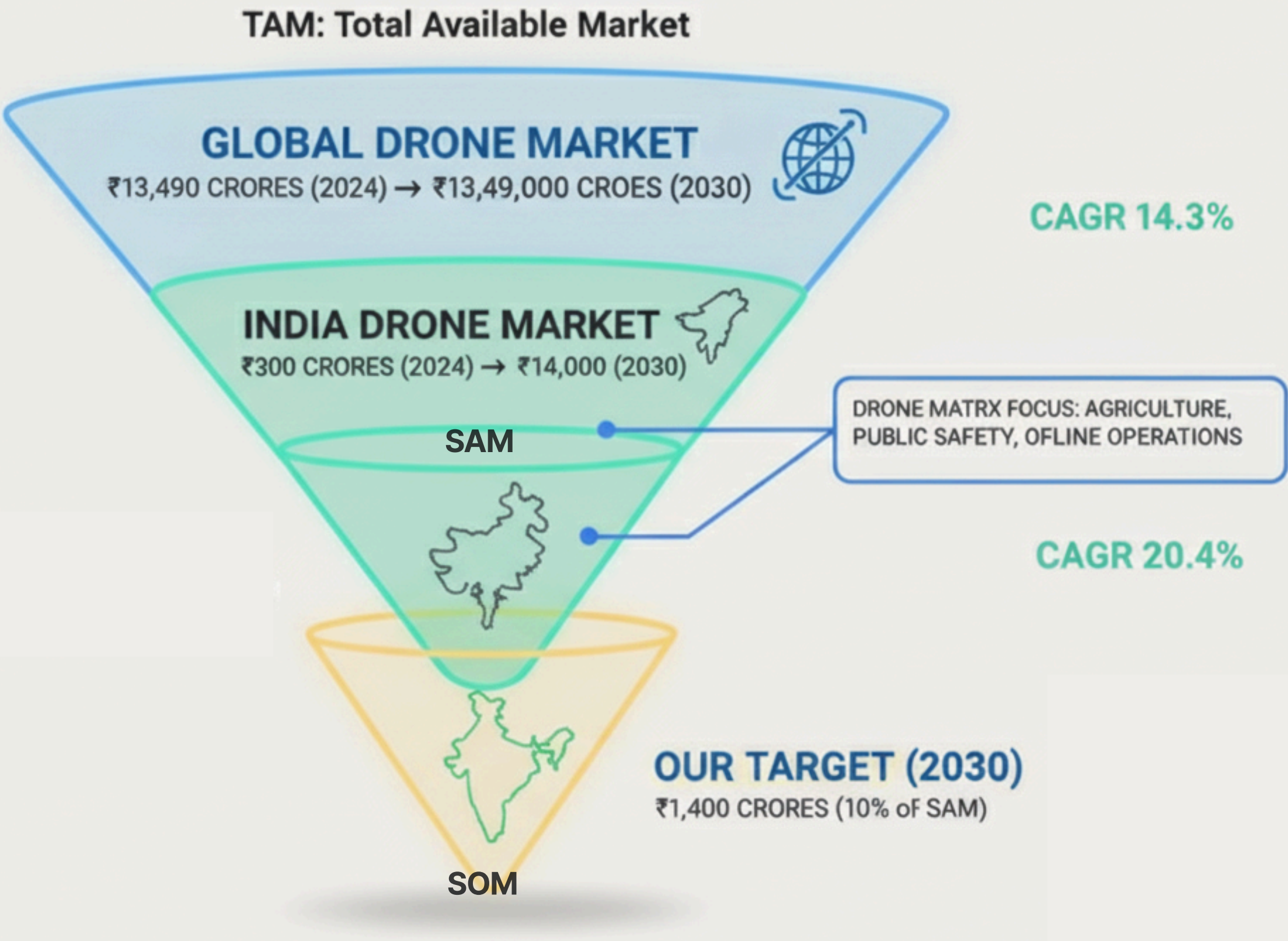
OPERATIONAL IMPACT ACROSS KEY SECTORS



These regions represent the highest operational impact zones where swarm enabled autonomy directly improves mission speed, coverage, and efficiency.

- SAR & DISASTER MANAGEMENT** : Auto divide search grid and parallel scanning, 2x quicker terrain assessment
- AGRICULTURE** : Synchronized multi drone spraying, coordinated crop monitoring
- INSPECTION & INFRASTRUCTURE** : Parallel inspection, reduce monitoring time and operational cost
- DEFENSE AND MILITARY** : Distributed mesh swarms, Low radio signature, continuous coverage, hard to detect

MARKET ANALYSIS



REVENUE MODEL

Open-Core + Usage + Enterprise

FREE, R&D OPEN CORE

- Up to 5 drones swarm control.
- Local, cloud free, MAVLink plugins.
- Object detection, NDVI View.
- Target: Labs, Developers, Universities.

PAY PER MISSION CREDITS

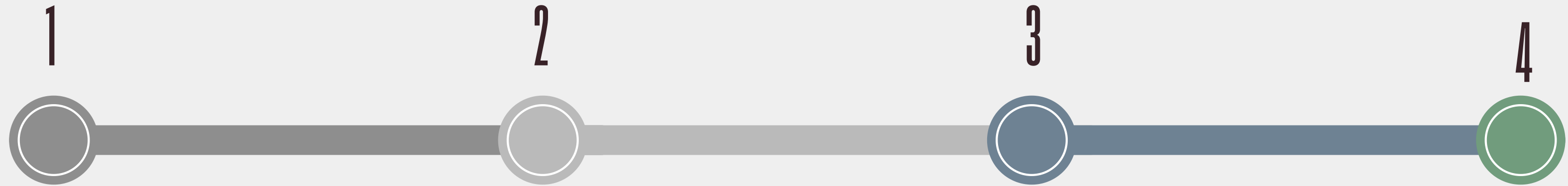
- Top up credits, priced by scale, duration, features.
- Includes analytics, maintenance dashboard
- Transparent, fair, per mission cost.
- Target: startups, operators, Project based teams.

ENTERPRISE LICENSE/SUBSCRIPTION

- Advanced AI, dynamic swarm path planning, private orchestration server.
- API integrations , SLA support.
- Revenue: annual subscription/license.
- Target: government, large enterprises.

Freemium drives adoption, mission credits scale revenue, enterprise licensing locks long term contracts.

TENTATIVE PLAN (NEXT 12 MONTHS)



(MONTHS 1–4) - CORE BUILD & SIMULATION

- Alpha : Multi UAV PX4/ArduPilot in SITL, AOI (coverage lanes).
- Basic UI (Plan, Live), telemetry, WebRTC video, data logging.
- Exit : 3–5 drones complete full missions in simulation.

(MONTHS 5–7) - FIELD ALPHA

- Real world test flights (1–2 drones) with geofence, HOLD, RTL, LAND.
- CBBA task split, safety checklist validation.
- Exit: End to end field mission.

(MONTHS 8–10) - BETA & PILOT

- Dynamic replanning, LoRa fallback, ORCA avoidance, offline maps.
- Partner pilot (NGO/Agri), collect KPIs (coverage per hr, ETA error).
- Exit: 2+ successful pilot missions.

(MONTHS 11–12) - V1.0 RELEASE

- Multi mission ops, role-based access, installer and support kit.
- Pricing, training, and onboarding finalized.
- Exit: v1.0 deployed, 2–3 live customer sites

REQUIRED CERTIFICATIONS AND APPROVALS

PHASE : 1

**GET
LEGAL**



- Incorporate your company (Pvt Ltd).
- Get PAN, TAN, GST.
- Register with DPIIT (Startup India) and MSME.

PHASE : 2

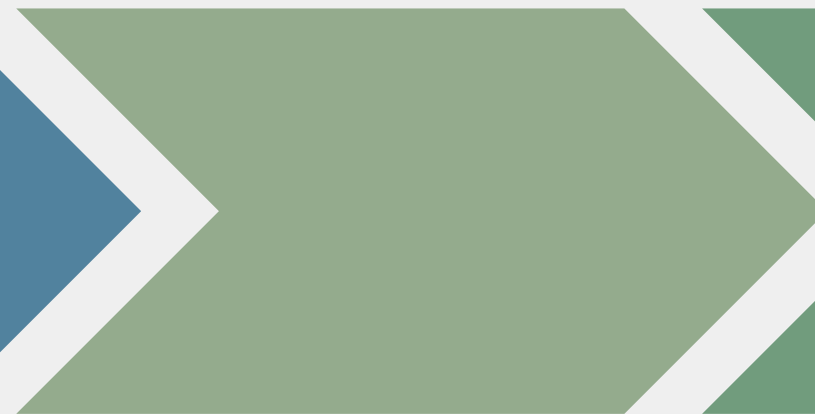
**GET
FLYING**



- Jointly apply for Type Certificate (TC) for this "Drone Matrix Powered" model.
- Simultaneously, build the Digital Sky (NPNT, NFZ) API integration.

PHASE : 3

**GET
TRUSTED**



- Start the process for ISO 9001 and ISO 27001.
- Begin the STQC certification process.
- Use Type-Certified drone to apply for a BVLOS/Swarm experimental permit with the DGCA.



EXPENSE BUCKET

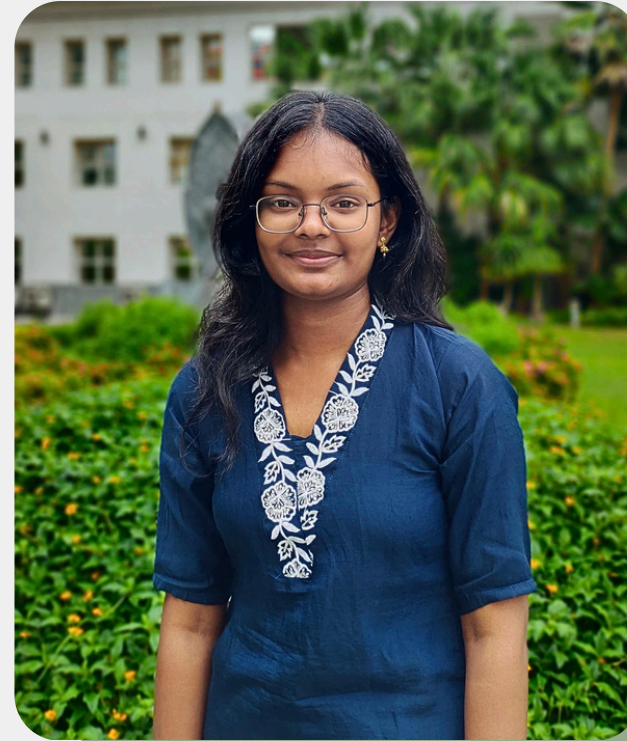
Category	Purpose / Activity	Amount (₹)	Expected Outcome
Technology & Development	Engineering & Design (Prototype build), Cloud/Comms Opex (TURN, storage, maps)	2,30,000	Working software, cloud services deployed
Hardware Development (Testing)	Testbed Drone & Payload (PX4/ArduPilot kit), Jetson/RPi kit, LTE modem, Wi-Fi AP, cables	2,40,000	Physical MVP hardware assembled
Software Testing & Compliance	Field Pilots, Training Compliance, Safety & Tools, Software Testing (cones, PPE, calibration)	70,000	User feedback gathered, safety protocols validated
IP & Legal	Patent, TYPE Certificate, ISO 9001, ISO 27001, SQTC, BVLOS	4,60,000	IP filed
	Total	10,00,000	

MEET OUR TEAM



RUDRA PATEL

- **REUDE Technologies** - UAV Navigation and Control Intern.
- **Karnavati Innovation and Incubation Centre** - Technical Event Management.



LITHIKA SARAVANAKUMAR

- **Developer Club** - Technical Head
- **GFG Campus Body** - Operations Head
- **Internship at Blackpapers** - Software Development.
- **Internship at Reude Technologies** - UAV Navigation and Control Intern



UTKARSH SHARMA

- **Developer Club** - Design and Creative Lead
- **GFG Campus Body** - Creative Head
- **Teachnook** - UI/UX Project
- **Freelance/Client Projects** - Front End & Brand Design

MENTOR
MR. ANIMESH KUMAR
AGGARWAL

- **Air Force Service (24 Yrs):** Two decades of service in the Air Force, demonstrating operational leadership, discipline, and advanced technical proficiency.
- **NFSU Forensic Expert (4 Yrs):** Expert in advanced scientific forensics and investigative analysis.
- **Corporate Experience (2 Yrs):** Specialized forensic consulting experience for the commercial sector.



Drone Matrix

THANK
YOU